

1) Electric Poles

In electrotechnic we are studying networks in *quasi stationary regimes*, this means that electric radiation is neglected and the parameters propagate instantaneously. The poles are connected via **ideal conductors**, in an **insulated space** with a **conservative electric field**. The main focus of the course will be with bipoles, but generally n-poles can be defined. These are the main components of an electric circuit. Let's see the 2 main properties for a bipole:

1. The current entering is the same as the one exiting, i.e. $\sum i_n = 0$
2. The voltage between the terminals is a difference of potential, i.e. the electric field outside is conservative

A pole is defined via it's characteristic equation of form:

$$\begin{aligned}v_{AB}(t) &= f(i(t)) \\ i(t) &= g(v_{AB}(t))\end{aligned}$$

1.1) Double bipoles

✓ TODO

1.2) Bipoles

As explained below bipoles are the main type of poles we are going to study, the specific ones are voltage and current sources, resistors, capacitors and inductances. For now let's just define the main properties of bipoles:

- Active bipole: $I = 0$, $V_{AB} \neq 0$
- Passive bipole: I and V_{AB} have the same sign
 - it is always an inert bipole: $I=0$, $V=0$
- Linear/Non-linear: self explanatory

Various bipoles can be connected in a network in the following ways:

- Series: $V = \sum V_i$, the same current flows in every bipole
- Parallel: $I = \sum I_i$, they all have the same difference of potential

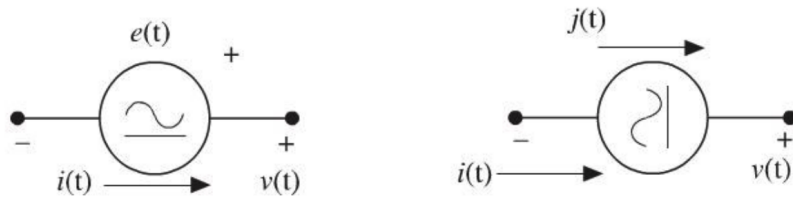
1.2.1) Ideal Voltage and Current sources

DC Regime

✓ DC REGIME

AC Regime

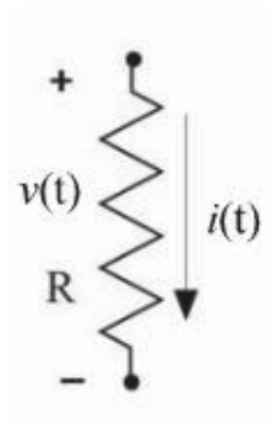
In **AC Regime** we can define these sources using a simple sinusoidal: $e(t) = E_M \sin(\omega t + \alpha)$,
 $j(t) = J_M \sin(\omega t + \alpha)$.



Graphical representation of the bipoles

1.2.2) Linear Resistor

Dc Regime



Resistor bipole

For a linear resistor we use the **Load Convention**. The characteristic equation becomes: $v_{AB}(t) = R i(t)$.

The inverse of a resistance is the **Conductance** $G = 1/R$ and then: $i(t) = G v_{AB}(t)$

For a resistor the power corresponds to Joule's law: $P = V_{AB}I = RI^2$

These are some useful theorems regarding resistors

THEOREM 1 (VOLTAGE DIVIDER)

In a series of n resistors, the difference in potential of the k -th resistor is:

$$V_K = R_k I = R_k \frac{V_{AB}}{R_{eq}} = V_{AB} \frac{R_k}{\sum R_i}$$

THEOREM 2 (CURRENT DIVIDER)

In a parallel of n resistors, the difference current of the k -th resistor is:

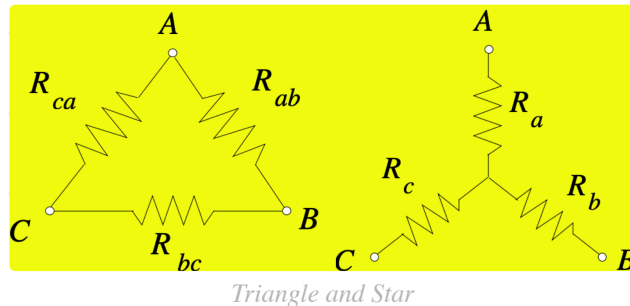
$$I_k = G_k V = G_k \frac{I}{G}$$

In the case of $n = 2$ the current divider can be easily written with R instead of G :

$$I_i = I \frac{R_j}{R_i + R_j}$$

THEOREM 4 (STAR/TRIANGLE TRANSFORMATION)

It is possible to transform the following shapes using these relations



$$R_{ab} = R_a + R_b + \frac{R_a R_b}{R_c} \iff R_a = \frac{R_{ab} R_{ac}}{\sum R_{ij}}$$

Ac Regime

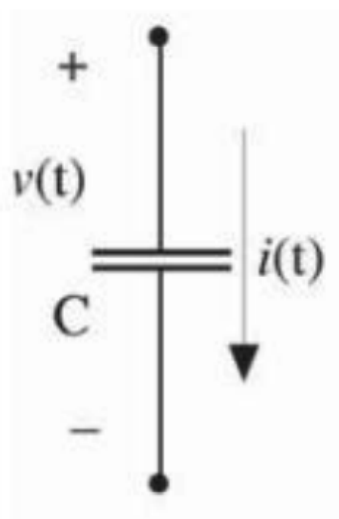
In **AC regime** a resistor keeps its characteristic function.

Being subject to alternating current it is subject to $v(t) = V_M \sin(\omega t + \alpha)$, and thus it imposes

$i(t) = I_M \sin(\omega t + \alpha)$. From here we can define $R = \frac{v(t)}{i(t)} = \frac{V_M}{I_M} = \frac{V}{I} = Z$

1.2.3) Capacitor

DC Regime



Capacitor bipole

A capacitor is made of two plates made of conducting material put near each other. The current entering the + side is different from the exiting one, since it **depends on the charges on each plate**. The capacitance $C = Q/V$ is a constant term since it is dependent on the geometric factor.

In order for a capacitor to be a bipole it must be imposed that the entering current is the same as the exiting one: $i_a(t) = i_b(t) \implies q_a(t) + q_b(t) = 0$. The bipole uses the **load convention** and has characteristic function $i(t) = dq(t)/dt = C \frac{dv(t)}{dt}$. From here $v(t) = v(0) + \frac{1}{C} \int_0^t i(\tau) d\tau$.

To find the characteristic function these steps were taken:

$$i(t) = d(CV)/dt = Cd(V)/dt + Vd(C)/dt = i(t)Cd(v(t))/dt$$

In a DC circuit it behaves like an open circuit.

In terms of power we have: $p = vi = vCv'$. from here we can define 2 works:

$$\text{Charge:} \quad W_C = \int_0^{V_M} p dt = \int C v dv = \frac{1}{2} C V_M^2 = \frac{1}{2} Q V_M = \frac{1}{2} \frac{Q^2}{C}$$

$$\text{Discharge:} \quad W_D = \int_{V_M}^0 p dt = \int C v dv = -\frac{1}{2} C V_M^2$$

THEOREM 5 (SERIES AND PARALLEL CONNECTION OF CAPACITORS)

The equivalent capacity is described as:

$$C_{eq} = \left(\sum \frac{1}{C_i} \right)^{-1}, \quad \text{in series}$$

$$C_{eq} = \sum C_i, \quad \text{in parallel}$$

Proof.

Proof for series connection:

$$\text{Since: } i = i_i = i_j \text{ and } v = \sum v_i \text{ it clearly becomes } v = q \sum \frac{1}{C_i} \implies C_{eq} = \frac{v}{q}^{-1} = \left(\sum \frac{1}{C_i} \right)^{-1}$$

Proof for parallel connection:

$$\text{Since: } i = \sum i_i \text{ and } v = v_i = v_j \text{ it clearly becomes } q = \sum q_i = \sum C_i v_i = v \sum C_i \implies C_{eq} = q/v = \sum C_i$$

□

✓ TODO: CHARGE AND DISCHARGE OF CAPACITOR

AC Regime

In **AC regime** a capacitor keeps it's characteristic function.

Being subject to alternating current it has a voltage $v(t) = V_M \sin(\omega t + \alpha)$ applied. This imposes that the current has the following amplitude: $I_M = \omega C V_M$ and is in anticipate quadrature with the voltage:

$$i(t) = C \omega V_M \cos(\omega t + \alpha) = \omega C V_M \sin(\omega t + \alpha + \pi/2). \text{ The symbolic representation is: } \bar{I} = \omega C \bar{V} e^{j\pi/2} = j\omega C \bar{V}.$$

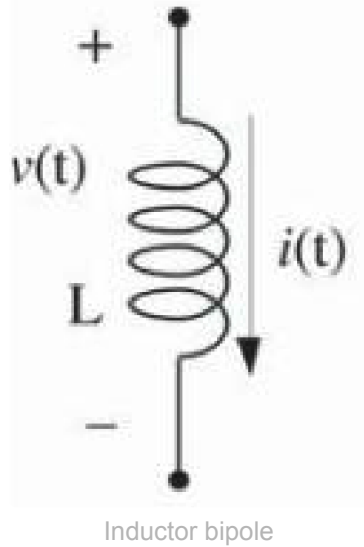
From here the impedance is: $\bar{Z} = 1/j\omega C = -j/\omega C = jX_C$ with $X_C = -1/\omega C$.

1.2.4) Inductor Bipole

DC Regime

An inductor works in such a way that it inhibits current changes in a circuit due to magnetic induction.

The bipole uses the **load convention** and in order for it to be a bipole it must hold the existence conditions: the current passing is



constant and since *externally of L the induction is null* the bipole has a difference of potential on it's terminals. Here are the characteristic equations:

$$v = L \frac{di}{dt}$$

$$i(t) = i(t_0) + \frac{1}{L} \int_{t_0}^t v dt$$

By recalling Lorentz law: $Li = \Phi_B$, by derivating $d\Phi_B/dt = L di/dt$ and by Faradays law: $d\Phi_B/dt = v$

A **real inductor bipole** is essentially a series of L and R: $v = v_L + v_R = Li' + Ri$

In power terms we define: $p = vi = Li'i$, then the infinitesimal work becomes: $dW = p dt = Lidi$ and the work done to charge is

$$W_C = \int_0^I Lidi = \frac{1}{2} LI^2$$

THEOREM 6 (SERIES AND PARALLEL CONNECTION OF INDUCTORS)

The equivalent inductance is:

$$L_{eq} = \left(\sum \frac{1}{L_i} \right)^{-1}, \quad \text{in parallel}$$

$$L_{eq} = \sum L_i, \quad \text{in series}$$

Proof.

For series connection:

All currents are the same, moreover $v = \sum v_i \rightarrow \sum L_i i' = i' \sum L_i$ then $v/i' = \sum L_i$

For parallel connection:

All voltages are the same, moreover $i = \sum i_i \rightarrow \sum i_i(0) + (\sum 1/L_i) \int_0^t v dt$, then

$L_{eq} = (i'/v)^{-1} = (\sum 1/L_i)^{-1}$ where the last step was made by deriving i

□

✓ CHARGE AND DISCHARGE OF INDUCTOR

AC Regime

In **AC regime** the inductor keeps it's characteristic function.

By applying a voltage $v = V_M \sin(\omega t + \alpha)$ we can find first di/dt , then by integration i .

$$di/dt = \frac{V_M}{L} \sin(\omega t + \alpha)$$

$$i = \frac{V_M}{L} \int \sin(\omega t + \alpha) dt = -\frac{V_M}{\omega L} \cos(\omega t + \alpha) = \frac{V_M}{\omega L} \sin(\omega t + \alpha - \pi/2)$$

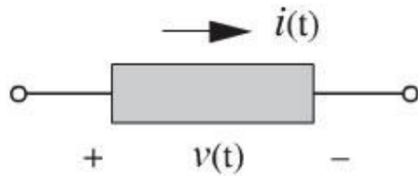
Then in symbolic representation: $\bar{I} = \bar{V} e^{-j\pi/2} = -j \frac{1}{\omega L} \bar{V}$. This is clearly in delay quadrature with $\bar{V}$

From here $\bar{Z} = \frac{\bar{V}}{\bar{I}} = j\omega L = jX_L$ with $X_L = \omega L$

2) DC Regime

2.1) Power and Conventions

An important parameter to define is power $p(t) = v_{AB}(t)i(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta Q}{\Delta T}$. It's sign is an important metric but in order to understand it's meaning we must first define 2 main conventions of bipoles:



Load convention in a generic bipole

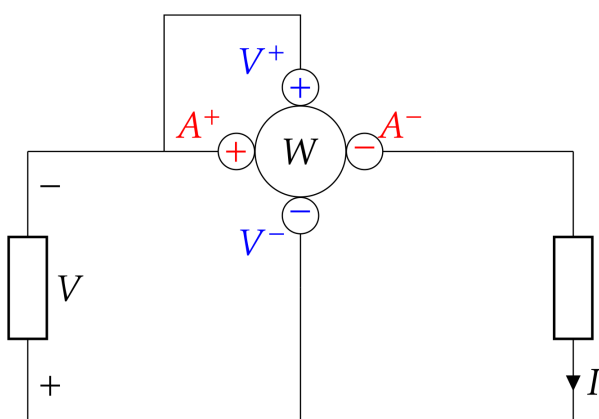
1. Load convention: I is positive if it enters on + side
2. Source convention: I is negative if it enters on + side

Now we can better understand the meaning of the power. A *source convention(generator)* will have a positive power if it is delivering power, while a *load convention(resistor)* will have a positive power when it is absorbing power. This will come useful when talking about: [Tellegen Theorem](#)

The work done to move an electron from point A to B is defined as:

$$\Delta \mathcal{L}_{AB} = \int_{\Delta t} p(t) dt$$

2.2) Measuring Devices



Wattmeter Example

In order to measure the power a wattmeter is used. This device is essentially a voltmeter and an ammeter combined.

In this example the current enters the A^+ side and thus it is positive. On the other hand the V^- side of the wattmeter faces the V^+ side of the bipole and thus the power measured will be: $P = -VI$.

3) AC Regime

In a periodic regime the time dependent quantities follow the following law: $a(t) = a(t + nT)$. We can then define the following values:

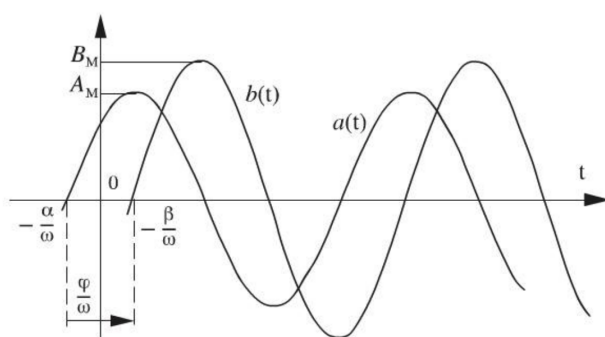
- Max value: $A_M = \max\{a(t)\}$
- Mean over one period: $A_m = \frac{1}{T} \int_0^T a(t) dt$
- Root Mean Square (RMS) : $A = \sqrt{\frac{1}{T} \int_0^T a^2(t) dt}$

In order to be AC the mean $A_m = 0$, and the period can be identified by two symmetric halves.

A sinusoidal function is the main function we study in this regime: $a(t) = A_M \sin(\omega t + \alpha)$, where the parameters have the following relation: $\omega = 2\pi f$. Now we can calculate the values:

- Max value: A_M
- Mean over one period: $A_m = 0$
- Mean over a half period: $A'_m = \frac{1}{T/2} \int_0^{T/2} A_M \sin(\omega t) dt = \frac{2A_M}{T} \left[-\frac{1}{\omega} \cos(\omega t) \right]_0^{T/2} = \frac{2}{\pi} A_M$
- Root Mean Square (RMS) : $A = \sqrt{\frac{1}{T} \int_0^T a^2(t) dt} = \frac{A_M}{\sqrt{2}}$
- Form factor: $K_f = \frac{A}{A'_m} = \frac{\pi}{2\sqrt{2}} = 1,11$

In the case of **2 isofrequency functions**, i.e. $a(t), b(t)$ with same frequency but different amplitude and phase the following relations might happen:



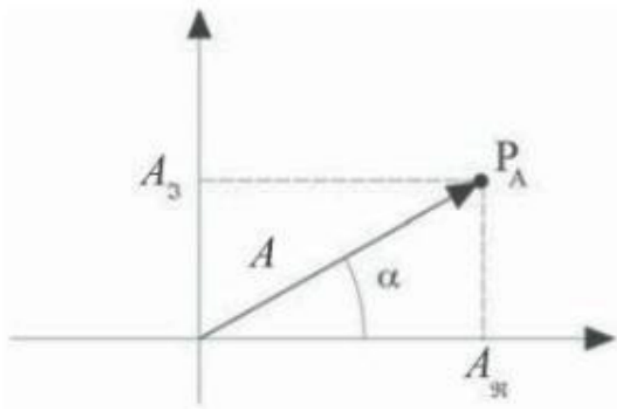
Two isofrequency functions

- In phase: $\alpha = \beta$
- Phase Opposition: $\alpha - \beta = \pi$
- Phase Anticipation: $\alpha > \beta$, in quadrature if $\alpha - \beta = \pi/2$
- Phase Delay: $\beta > \alpha$, in quadrature if $\beta - \alpha = -\pi/2$

REMARK 7 (ARCTANGENT ANGLE ERROR)

When calculating the phase of the sinusoidal representation starting from the complex representation it is important to check the sign of the real part. **If the real part is smaller than 0 it is necessary to add π to the resulting phase**

3.1.1) Phasor Representation



Graphical representation of a phasor

The phasorial representation shows $a(t)$ as a vector of modulus A_M rotating counterclockwise with angular velocity ω and starting angle α . The projection of the phasor on the x and y axis show respectively the cosine and sine function.

3.1.2) Symbolic representation

More useful is the symbolic representation. Usually a sinusoidal is expressed in the form:

$$\bar{A} = \sqrt{2}A_{\text{eff}}e^{j\alpha}e^{j\omega t} = A_M e^{j\alpha}e^{j\omega t} = A_M e^{j(\omega t + \alpha)}$$

A simple proof consists of taking the phasorial representation in complex terms:

$\bar{A} = A_M \cos(\omega t + \alpha) + jA_M \sin(\omega t + \alpha)$. By using $\cos(a + b) = \cos(a)\cos(b) - \sin(a)\sin(b)$ and $\sin(a + b) = \sin(a)\cos(b) + \sin(b)\cos(a)$ we get:

$$\begin{aligned} & A_M(\cos(\omega t)\cos(\alpha) - \sin(\omega t)\sin(\alpha) + j\sin(\omega t)\cos(\alpha) + j\cos(\omega t)\sin(\alpha)) = \\ & A_M(\cos(\alpha)(\cos(\omega t) + j\sin(\omega t)) + j\sin(\alpha)(\cos(\omega t) + j\sin(\omega t))) = \\ & A_M(\cos(\alpha) + j\sin(\alpha))(\cos(\omega t) + j\sin(\omega t)) \end{aligned}$$

with the term $A_M(\cos(\alpha) + j\sin(\alpha))$ being $\bar{A}$ at $t = 0$.

3.1.3) Impedance and admittance

Define a passive bipole using load convention(resistor, capacitor, inductance) in a AC circuit, we define the impedance as : $\bar{Z} = \frac{\bar{V}}{\bar{I}} = Z e^{j(\alpha - \beta)} = Z e^{j\phi}$. This complex operator shows the tendency of the bipole to resist the change of current passing through it

The admittance is the reciprocal of the inductance

It is possible to define a parallel and series connection of all passive bipoles and find their equivalent impedance:

$$\begin{aligned} \bar{Z}_{\text{series}} &= R_{\text{series}} + jX_{\text{series}} = \sum \bar{Z}_i \\ \bar{Z}_{\text{parallel}} &= (R_{\text{parallel}} + jX_{\text{parallel}})^{-1} = (\sum 1/\bar{Z}_i)^{-1} \end{aligned}$$

Here is a quick recap of impedance for the passive bipoles we are studying:

	Z	X	phase with $\bar{V}$
R	R		in phase
C	jX_C	$-1/\omega C$	anticipation quadrature
L	jX_L	ωL	delay quadrature

3.1.4) Power

☒
TODO

3.1.5) Resonance

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TODO

4) Circuits

4.1) Theorems

THEOREM 8 (TELLEGEN THEOREM)

In a circuit in DC regime, the total absorbed power is the same as the total supplied power

$$\sum_{k=1}^{R_n} p_{abs_k}(t) = \sum_{k=1}^{E_n} p_{sup_k}(t)$$

It also holds in a circuit in AC Regime

$$\sum_{k=1}^{R_n} p_{abs_k}(t) = \sum_{k=1}^{E_n} p_{sup_k}(t)$$

THEOREM 9 (NON AMPLIFICATION THEOREMS)

In a circuit made only of bipoles with one single source bipole, at any time t , the following theorems are valid:

Voltage non amplification:

The module of the supplied power is greater or equal to the voltage on any other bipole.

Current non amplification:

The module of the current supplied by the source is always greater or equal to the current on any other bipole

Proof.

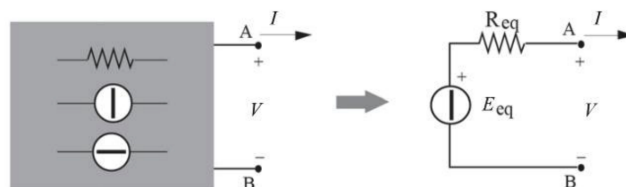
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The following theorems hold in DC Regime:

THEOREM 10 (THEVENIN THEOREM: EQUIVALENT VOLTAGE GENERATOR)

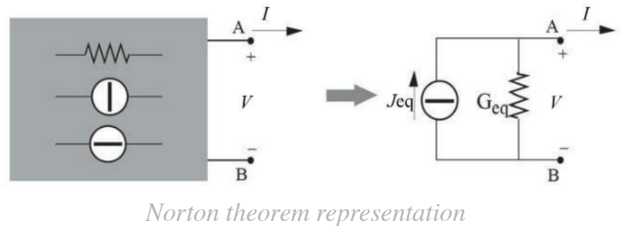
The behaviour of any bipole network, in terms of voltage at the terminals A , B is equivalent to a series constituting of E_{eq} , R_{eq}



Thevenin theorem representation

with $E_{eq} = V_0$ the tension between the terminals A , B and R_{eq} the equivalent resistance with all generators turned off. Keep in mind $R_{eq} = \frac{V_0}{I_{cc}}$

The behaviour of any bipole network, in terms of voltage at the terminals A, B is equivalent to a parallel constituting of J_{eq}, G_{eq}



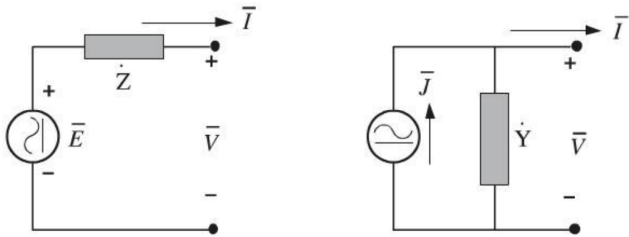
Norton theorem representation

With $J_{eq} = I_{cc}$ and $G_{eq} = \frac{I_{cc}}{V_0} = \frac{1}{R_{eq}}$

The equivalent version of these theorems that holds in AC regime is:

THEOREM 12 (EQUIVALENT SYMBOLI CCURRENT / VOLTAGE SOURCE)

The behaviour of any bipole network in AC regime can be written as either an equivalent voltage source and impedance, or an equivalent current source and an admittance.



Representation of the theorem

4.2) Resolution methods

4.2.1) Kirchhoff laws

DC Regime

In a circuit identify a graph
When using Kirchhoff laws in a circuit you have to grate 2 graphs:

- The main graph is connected to all nodes only once
- The second graph is made of all missing branches

then by defining l as the number of branches, and n the number of nodes a number of $n - 1$ branches equations and $l - (n - 1)$ loop equations are needed. Recall the 2 seconds law that state: In a closed and subset of branches cut by a surface (or a node) the sum of all currents is null, and for each mesh/loop the sum of all voltages is null. It is clear that the $n - 1$ nodes equation will be of the form $\sum i_{in} = \sum i_{out}$ while the loop equations will have the following form:

	V +	V -
R	I same as loop	I opposite as loop
E	loop enters - side	loop enters + side

	V +	V -
J	J opposite as loop	J same as loop

AC Regime

Kirchhoff also holds in AC Regime:

- $\sum \bar{I}_h = 0$ on an insieme di taglio (node)
- $\sum \bar{V}_h = 0$ on a branch

Never use the abs. value of a phasor!

4.2.2) Superimposition

Works for both DC and AC regimes.

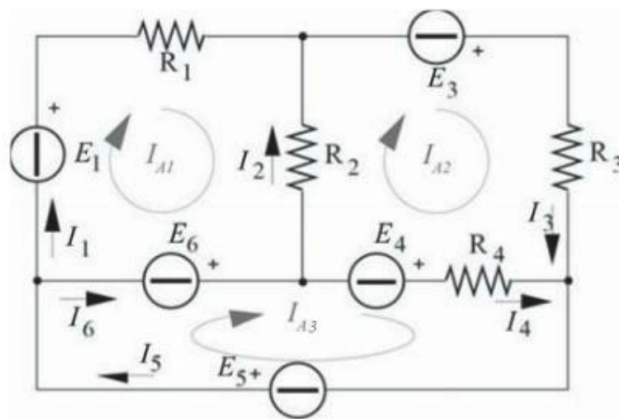
Given a **linear circuit** define the current direction for every branch. Now consider the circuit one generator at a time to find R, V while still keeping in mind the originally defined currents to keep the signs correct. After having found I, V for every source sum them to find I, V of the circuit in every branch. Unused sources become short circuit if they were generators, current generators become open circuits. Clearly I is positive if the current of the generator matches the one defined at the start. V is positive if the current touches first the "high" potential side

4.2.3) Loop current analysis

It is a way to solve a circuit using loop currents that implicitly satisfy the $n - 1$ LKC equations. Given a circuit it is necessary to define a loop direction for every loop (possibly all same orientation) I_{A_n} that may or may not coincide with the current direction defined for that branch. Every branch can be expressed as a sum of the adjacent loop currents. This is possible for all branches except the one with a current generator.

The expression for the n-th loop is made of many terms:

- Sum of all resistances times I_{A_n}
- Subtraction of RI in every branch with it's loop current.
- Are equal to the sum of the generators with respect to I_{A_n}



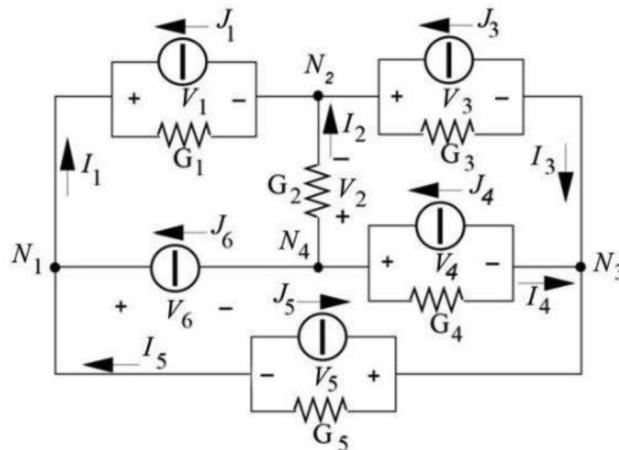
Example

In this example $I_2 = I_{A_2} - I_{A_1}$, moreover

$$\begin{cases} (R_1 + R_2)I_{A_1} - R_2I_{A_2} = +E_1 - E_6, & \text{loop } A_1 \\ (R_2 + R_3 + R_4)I_{A_2} - R_2I_{A_1} - R_4I_{A_4} = +E_3 - E_4, & \text{loop } A_2 \\ R_4I_{A_4} = +E_4 + E_5 + E_6, & \text{loop } A_3 \end{cases}$$

4.2.4) Node voltage analysis

This resolution method is equivalent to the but with voltages. This method satisfies implicitly the $l - (n - 1)$ KVL equations. To apply this method it is necessary to put a potential of a node to 0 in order to assume some reference potential, since the circuit is, by definition, [electrically floating](#). Now all the potentials can be expressed as a difference of potential. In this case every voltage can be expressed as: $V_n = U_+ - U_-$. While the equation for each node is of form: $(\sum G_i) - \sum G_i U_i = \sum J_{\text{enter}} - \sum J_{\text{exit}}$



Example

Let's set $U_4 = 0$, for example, $V_1 = U_1 - U_2 =$. Now:

$$\begin{cases} (G_1 + G_5)U_1 - G_1U_2 - G_5U_3 = J_1 - J_5 + J_6, & \text{for } N_1 \\ (G_1 + G_2 + G_3)U_2 - G_1U_1 - G_3U_3 = -J_1 + J_2 + J_3, & \text{for } N_2 \\ (G_3 + G_4 + G_6)U_3 - G_5U_1 - G_3U_2 = J_3 - J_4 + J_5, & \text{for } N_3 \end{cases}$$

Then, recalling Ohm's law, $I = GV$. With $V = U_+ - U_-$

If there are G and J in series, G doesn't appear in the equation

4.2.5) Substitution Principle

DC Regime

In a circuit with a **series connection of J, R can be substituted with just J** . This doesn't change V, R, J on the branch and thus it is equivalent

Analogously, a **parallel connection of E, R can be substituted with a single branch of E** This doesn't change V, R, J on the branch and thus it is equivalent.

To gather informations on the branch it is necessary to transform it back.

In the parallel connection of E, R the current flows in opposite directions
In the series connection J, R the voltage is different between the bipoles

In a circuit it is possible to substitute a **parallel connection of J, R in a series connection of E, R** with $E = RJ$.

In a circuit it is possible to substitute a **series connection of E, R in a parallel connection of J, R** with $J = E/R$.

To gather informations on the branch it is necessary to transform it back.

In the initial parallel connection of J, R the current flows in opposite directions
In the initial series connection E, R the voltage is different between the bipoles

AC Regime

It essentially holds the same but by just considering every passive bipole, not just R